

Jay Rao

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EDUCATION

Northwestern University
Bachelor of Arts (B.A.) in **Computer Science**

Expected December 2027
Concentration in **Artificial Intelligence**

SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js, React, PyTorch, NumPy, Pandas, Playwright

Tools & Platforms: AWS, Terraform, Docker, Git, GitHub, Linux/Bash, Render, Vercel

EXPERIENCE

Founding Engineer | GridGate LLC

Feb. 2026 - Present

- Sole engineer on GridGate Queue Analysis v1.0, a full MISO queue intelligence system: 19-source data lake (20+ scrapers), 9-module feature pipeline, and two CatBoost models trained on 3,700+ historical projects
- Built a two-stage ML pipeline: success/withdrawal classifier (ROC-AUC 0.93, 85% accuracy) and 5-class withdrawal phase classifier (74% accuracy, one-vs-rest ROC-AUC 0.94), both with SHAP explainability
- Automated collection of 300,000+ state/county regulations and meeting minutes PDFs across 15 states and 1000+ MISO counties via custom data scraping scripts and Claude agent discovery for non-standard sites
- Engineered a 6-tier developer-matching pipeline to map 3,700+ projects to their parent developers with 83% precision, using document-grounded lookups, Claude research, and multi-stage evidence verification

Technology Manager | iloveyou LLC

Feb. 2026 - Present

- Lead developer of an early-stage consumer social app, managing 3 junior developers across sprint planning, task assignment, and code review, to a hard August 12th launch
- Built the backend from scratch in Node.js/Express on PostgreSQL, including JWT auth with refresh-token rotation, email and phone verification (Amazon SES, Twilio Verify), APNs push notifications, and REST APIs
- Implemented the full iOS frontend in SwiftUI (MVVM): onboarding, fruit community assignment, profile creation, and all core functionality, including feed, messaging, events, and groups
- Engineered the AWS infrastructure in Terraform (VPC, ECS Fargate, ALB, RDS, S3, CloudFront, SES, Secrets Manager, IAM) with remote state and DynamoDB locking, plus CI/CD via GitHub Actions

Computer Science Educator | Northwestern TIDAL & tilt labs

Jan. 2026 - Jun. 2026

- Co-led two fifth-grade classrooms in the Evanston/Skokie School District (D65) each week, guiding 40+ students through creative, project-based Python curriculums using TunePad and SportSense

PROJECTS

game.sense | *Python, PyTorch, GNN, LSTM, carball, PyQt6*

- Creating a spatial intelligence platform for real-time POV coaching and post-game analysis using predictive modeling and multi-entity tracking, validated in a physics-based environment (Rocket League)
- Built a self-reinforcing data loop where post-game replay files are parsed into ground-truth entity states, training a GNN + LSTM model to infer hidden entity positions from a geometrically-masked first-person perspective across 1000s of matches for a live game state Windows overlay via WebSocket

Landscape Guessr | *Ruby on Rails, JavaScript, PostgreSQL, PostGIS, Stimulus, Tailwind CSS, Gemini API*

- Co-developed an interactive geographical guessing game in a 4-person team, allowing users to drop pins on a MapTiler-powered interface to estimate the locations of custom AI-prompted Wikimedia image sets
- Built a real-time multiplayer backend supporting live matchmaking, asynchronous challenges, practice mode with AI hints, and global leaderboards, integrating AWS Rekognition for automated image moderation

vibes.fm | *JavaScript, React, Node.js, AWS, MySQL, Transformers.js, OAuth 2.0*

- Engineered a full-stack AI playlist generator that encodes track metadata and lyrics into vector embeddings using a local sentence-transformer model, with vector cache and cosine similarity search to match natural language user “vibe” prompts to songs, exporting matched playlists directly to Spotify via PKCE OAuth 2.0

nuPython | *C, C++, STL*

- Engineered a complete interpreter, memory management (RAM), and GDB-style debugger for nuPython across 2,000+ lines of C/C++, supporting integers, floats, strings, booleans, pointers, and None types